

AI-AUGMENTED BACKEND DEVELOPMENT

AI BlogNest API

7. Project Documentation Phase

Technical, API, architecture, database, security, and AI documentation

1. Documentation Objective

The documentation phase records the system's purpose, requirements, architecture, database design, API behavior, security model, AI integration, testing evidence, and usage instructions so that the project can be understood and maintained independently.

2. Project Overview

AI BlogNest API is a secure, modular REST backend for blog creation and content management. The sample describes a Node.js and Express.js backend using MongoDB with Mongoose, JWT authentication, role-based access control, bcrypt password hashing, validation, sanitization, rate limiting, structured logging, blog CRUD, categories/tags, comments, media metadata, search, and AI services for blog generation and summarization. ■filecite■turn0file0■L5-L20■

3. Problem and Solution

The documented case study identifies difficulties in organizing drafts, categories, tags, analytics, approvals, comment moderation, media handling, and collaborative permissions. BlogNest addresses these through a centralized backend with permission controls and dedicated workflow services. ■filecite■turn0file0■L22-L48■

4. Architecture Documentation

Document the API gateway, authentication middleware, role matrix, business services, database interface, and AI flow. The sample explicitly presents these as the principal layers of the server architecture. ■filecite■turn0file0■L65-L82■

5. Database Documentation

Document User and Post fields, relationships, validation rules, and timestamps. The sample identifies User and Post entities and describes User-to-Post, User-to-Comment, and Post-to-Comment relationships. ■filecite■turn0file0■L86-L108■

6. API Documentation

- Authentication endpoints.
- Blog CRUD endpoints.
- Category and tag endpoints.
- Comment and moderation endpoints.
- Search endpoints.

- AI generation endpoint: `/api/ai/generate-blog`.
- AI summarization endpoint: `/api/ai/summarize`.

7. Security Documentation

- JWT authentication.
- bcrypt password hashing.
- Role-based route guarding.
- Input validation and sanitization.
- XSS and NoSQL injection-related safeguards.
- Rate limiting.
- Centralized error handling.
- Structured logging.

8. User and Developer Guidance

Documentation should explain the project setup, environment requirements, database connection, server startup, API testing process, role permissions, and AI workflow. The sample includes project setup, npm initialization, server structure, MongoDB configuration, API testing, and demonstration references.

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9. Documentation Deliverables

- Project overview and objectives.
- Software and hardware requirements.
- Architecture and MVC description.
- Entity and relationship description.
- API endpoint reference.
- Security and authorization rules.
- AI integration explanation.
- Testing results/evidence.
- Demo and code references.